Immunity Profile Methods Updates

Technical Specification, Assumptions, and Open Questions

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Abstract

This document describes methodological updates to the immunity profile and reporting rate estimation framework originally developed in Thakkar et al. [2025] for measles in Southern Nigeria. That work introduced a cohort-based immunity accounting model in which routine immunization, supplementary immunization activities (SIAs), and natural infection compete chronologically to confer immunity, with the residual infection probability $P_b(I)$ determined by a full-immunity normalization constraint.

In adapting this framework to Pakistan, we have made several updates:

- **Sub-national SIA targeting** (Part I, Section 3): SIA eligibility is now province-specific rather than national, with coverage computed against the targeted-area eligible population.
- **Unified chronological processing** (Part I, Section 4): All vaccination events (MCV1, MCV2, SIAs) are processed in a single chronological pass per semimonthly sub-cohort, correctly handling SIAs that fall between MCV1 and MCV2 ages and incorporating birth seasonality.
- **Updated MCV2 formulation** (Part I, Section 5): The original recursion treated MCV1 and MCV2 receipt as independent, potentially overestimating MCV2-derived immunity. The new formulation conditions on MCV1 receipt, treating MCV2 as a second dose.

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Part I
Immunity Profile Construction

1 Overview

Each province-level immunity profile estimates, per birth cohort b, the probability that a randomly chosen individual derives immunity from each possible source: routine immunization (MCV1, MCV2), supplementary immunization activities (SIAs), or natural infection. These probabilities are assumed to sum to one (the full-immunity assumption, Section 6), meaning that those who are not vaccinated are assumed to eventually get infected.

The key model inputs (per province and birth semimonth t) are:

- vaccination coverage by birth semimonth: province-level MCV1 and MCV2 coverages $C_1(t)$ and $C_2(t)$[1], and SIA-specific coverages (see Section 3);
- vaccine efficacies (ε_1, ε_2, ε_{SIA});
- semimonthly births $B(t)$[1];
- observed cohort age-at-infection distributions based on semimonthly case estimates[1].

The desired outputs are:

- $P(s = s' \mid b)$ for each immunity source s' and birth cohort b;
- model-implied cohort age-at-infection distributions $P(a \mid s = I, b)$.

Combining $P(s = I \mid b)$, $P(a \mid s = I, b)$, and $B(b)$ across birth cohorts gives us the following additional outputs:

- model-implied expected infections by year and age, $I(a, y)$;
- model-implied year age-at-infections distributions $P(a \mid s = I, y)$.

Finally, comparing expected infections by year to reported cases allows us to estimate

- the reporting rate $r(y)$.

To make cohort-based and year-based distributions more easily visually distinguishable, we will often write $P_b(a \mid I) := P(a \mid I, b)$ and $P_y(a \mid I) := P(a \mid I, y)$. We will also use the shorthand $P_b(I) := P(s = I \mid b)$ for the lifetime infection probability in cohort b.

[1] Estimated using Thakkar et al. [2025]'s methods (not discussed in this document).

2 Notation

Symbol	Meaning
b	birth cohort (year)
t	birth sub-cohort (semimonth)
A	assumed maximum age at infection (we take $A = 25$)
$a \in \{0, \ldots, A\}$	age in integer years
a_{sm}	age in semimonths (float)
$y = b + a$	calendar year
$P_b^{\text{obs}}(a \mid I)$, $\hat{P}_b(a \mid I)$	observed and model-implied cohort-level age-at-infection PMF
$P_y^{\text{obs}}(a \mid I)$, $\hat{P}_y(a \mid I)$	observed and model-implied year-level age-at-infection PMF
$B(b)$ or $B(t)$	births in year b or semimonth t
$P_b(I)$	lifetime probability of infection for cohort b
$F(a_{\text{sm}}, b)$	cumulative prob. of infection before age a_{sm} in cohort b
$S(a_{\text{sm}}, b)$ or $S(a_{\text{sm}}, t)$	susceptible fraction of cohort b or sub-cohort t at age a_{sm}
$I(a, y)$	number of infections at age a in year y
$I(y)$	number of infections in year y
$r(y)$	reporting rate in year y
$n_b(a)$	number of observed cases from cohort b at age a
n_b	total number of observed cases from cohort b
$n_y(a)$	number of observed cases in year y at age a
n_y	total number of observed cases in year y
$\varepsilon_1, \varepsilon_2, \varepsilon_{\text{SIA}}$	vaccine efficacies (MCV1, MCV2, SIA)
s	potential immunity source (infection, MCV1, MCV2, SIAs)
$C_s(b)$ or $C_s(t)$	coverage of vaccination event s for cohort b or semimonth t

3 Sub-national SIA targeting

In Pakistan, many SIA campaigns target specific provinces rather than the entire country. The original model treated all SIAs as national: eligibility was determined solely by age at campaign time, and all age-eligible children were assumed to receive doses. If this same approach were applied to sub-national SIAs, the impact of these SIAs on non-targeted provinces would be overstated, and vaccine-derived immunity would be overestimated.

Refined approach. SIA eligibility is now encoded at the province×semimonth level. For each campaign s', the SIA calendar specifies a list of targeted provinces. The semimonthly dataset now carries a 0/1 indicator column for each campaign, where 1 indicates that a child born in that province at that semimonth is both (i) age-eligible and (ii) in a targeted province:

$$\text{elig}_{s'}(t, \text{prov}) = \begin{cases} 1 & \text{if prov} \in \text{targets}(s') \text{ and } t \in \text{DOB window}(s') \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

where the DOB window is derived from the campaign date and age range.

For province-level immunity profiles, only campaigns targeting that province contribute. For the national profile, all campaigns contribute with age-eligibility only (province targeting is ignored), reflecting the population-average impact.

Coverage computation. SIA "coverage" is computed as doses delivered divided by eligible population. For sub-national campaigns, the eligible population is the sum of births in the DOB window across targeted provinces only:

$$C_{s'} = \frac{\text{doses}(s')}{\sum_{\text{prov}\in\text{targets}(s')} \text{elig}_{s'}(\text{prov})}, \tag{2}$$

where $\text{elig}_{s'}(\text{prov}) = \sum_t B(t) \cdot \text{elig}_{s'}(t, \text{prov})$ is the eligible birth count for that province. This gives a higher (and more accurate) coverage rate for sub-national campaigns than dividing by the national eligible population.

It is worth emphasizing that by "coverage," we mean the portion of the age-and-province-eligible population (implied by our semimonthly births estimates) that there were doses available for. This is distinct from how SIA coverage is often defined, which is doses given divided by the *campaign-defined* target population. This is an important distinction, since district-specific campaigns could have high campaign-defined coverage, while only reaching a small portion of the age-eligible kids in that province.

Assumption: equal-likelihood of SIA dose receipt

We assume that all eligible children are equally likely to receive SIA doses, regardless of prior immunization/infection history, and regardless of which eligible province they are in. We make this assumption because of two main challenges:

- We don't yet have a good way to estimate RI and SIA dependence, so we assume for now that they are independent. This may not hold perfectly: children harder to reach for routine immunization may also be harder to reach by SIAs, or children who are known to be immune from RI may be less likely to participate in SIAs.
- Our input SIA data does not have information about how much each SIA prioritized each province, so we can't easily say "SIA X gave 70% of its doses to Punjab and 30% of its doses to Sindh," we just know that the SIA included Punjab and Sindh (and sometimes it doesn't even say which provinces were included).

4 Unified chronological processing

Rather than computing MCV1 and MCV2 immunity separately and then iterating over SIAs, we process *all* vaccination events in a single chronological pass per sub-cohort. This ensures that the susceptible fraction at each event correctly accounts for all prior events, regardless of their type.

4.1 Per-semimonth vaccination timelines

For each semimonth t within birth year b, we have estimates of the following (using methods from Thakkar et al. [2025]): the number of births $B(t)$, MCV1 vaccination coverage $C_1(t)$, and MCV2 vaccination coverage $C_2(t)$. For children born in semimonth t, the complete vaccination timeline is:

- **MCV1** at calendar time $t + 9$ months, with coverage $C_1(t)$ and efficacy ε_1;

- **MCV2** at calendar time $t + 15$ months, with coverage $C_2(t)$ and efficacy ε_2;
- **Each SIA** s' at its campaign date $t_{s'}$, with population-level coverage $C_{s'}$ and efficacy ε_{SIA}, if the child is age-eligible.

These events are sorted chronologically, depending on the birth semimonth: for a child born in January, MCV1 (October) may precede an SIA in November of the same year; for a child born in September, the same SIA may precede MCV1 (June of the following year).

Future work: modeling MCV1 and MCV2 receipt ages as distributions

In the above approach, we assume every child who receives an MCV1 dose receives their dose at exactly 9 months of age, and similarly for MCV2 at 15 months of age. In practice, these receipt ages are likely closer to 9 ± 2 months and 15 ± 2 months. In future work, we could model MCV1 and MCV2 receipt ages as compact distributions centered at 9 and 15 months of age, respectively.

4.2 Initializing cohort-based age-at-infection distributions

The process for initializing $P_b(a \mid I)$ from $P_b^{\text{obs}}(a \mid I)$ will be described in Part **??**.

4.3 Cumulative infection at fractional ages

Each vaccination event occurs at a (fractional) age $a_{\text{sm}} = (t_s - t)/365.25$. We obtain age-at-infection distributions at these fractional ages by linearly interpolation of the integer-age distributions $P_b(a \mid I)$:

$$P_b(a_{\text{sm}} \mid I) = (1 - \alpha)\, P_b(\lfloor a_{\text{sm}} \rfloor \mid I) + \alpha\, P_b(\lceil a_{\text{sm}} \rceil \mid I), \qquad \alpha = a_{\text{sm}} - \lfloor a_{\text{sm}} \rfloor. \tag{3}$$

The cumulative infection probability at age a_{sm} can then be written as follows:

$$F(a_{\text{sm}},\, b) = P_b(I)\, P_b(\text{age} \le a_{\text{sm}} \mid I), \tag{4}$$

where $P_b(I)$ starts as an unknown but is solved for as described in Section 6.

4.4 Processing events in order

For children born in semimonth t within birth year b, let $s_1 \prec s_2 \prec \cdots \prec s_M$ be the chronologically ordered vaccination events. We maintain a running total of vaccine-derived immunity $V_{\text{cum}}(t)$, initialized at zero. At each event s_j:

(i) Compute the cumulative infection probability $F(a_{s_j}, b)$ at the event age.

(ii) Compute the susceptible fraction: $S(a_{s_j}, t) = 1 - V_{\text{cum}}(t) - F(a_{s_j}, b)$.

(iii) Compute new immunity from this event (see below).

(iv) Update $V_{\text{cum}}(t)$ += new immunity.

The new immunity depends on the event type:

MCV1. Assuming that there is no vaccine-derived immunity prior to a_{MCV1}, MCV1-derived immunity can be expressed as follows:

$$P(s = \text{MCV1} \mid t) = \varepsilon_1 \, C_1(t) \, S(a_{\text{MCV1}}, t). \tag{5}$$

MCV2 (updated formulation). We assume that MCV2 receipt guarantees that MCV1 was received, meaning MCV2-derived immunity requires MCV1 failure (see Section 5 for more detail):

$$P(s = \text{MCV2} \mid t) = \varepsilon_2 \, C_2(t) \, (1 - \varepsilon_1) \, \left(1 - F(a_{\text{MCV2}}, b) - V_{\text{cum}}^{\text{SIA}}(t)\right), \tag{6}$$

where $(1 - \varepsilon_1)$ is the probability that MCV1 failed to seroconvert, and $V_{\text{cum}}^{\text{SIA}}(t)$ is the running total of SIA-derived immunity pre-MCV2 age.

SIAs. For SIA s', the probability of receipt is assumed to be independent of RI, so the following general formula (analogous to (5)) applies:

$$P(s = s' \mid t) = \varepsilon_{\text{SIA}} \, C_{s'}(t) \, S(a_{s'}, t). \tag{7}$$

4.5 Aggregation to birth-year resolution

After processing all events for each semimonth, per-source immunity fractions are aggregated to birth-year level by weighting by births:

$$P(s = s' \mid b) = \frac{\sum_{t \in b} B(t) \; P(s = s' \mid t)}{\sum_{t \in b} B(t)}. \tag{8}$$

This naturally incorporates birth seasonality: if most births in cohort b occur in months where MCV1 falls before a particular SIA, the aggregated profile reflects this.

5 MCV2: updated formulation

MCV2 is a second dose in the routine schedule, administered at approximately 15 months. We assume that receiving MCV2 *guarantees* having received MCV1, which influences the likelihood of prior immunity.

The original (incorrect) formulation. Earlier work — which also assumed for the purposes of estimating MCV2-derived immunity that there is no SIA-derived immunity between MCV1 and MCV2 ages — applied the general SIA recursion (7) directly to MCV2:

$$P(s = \text{MCV2} \mid b) = \varepsilon_2 \, C_2(b) \, (1 - F(a_{\text{MCV2}}, b) - P(s = \text{MCV1} \mid b)) \, . \tag{9}$$

The RHS of this formula can be interpreted as (probability MCV2 dose seroconverts)∗(probability MCV2 dose is received)∗(probability of no prior immunity at MCV2 age).

The issue is that this last term should really be "probability of no prior immunity at MCV2 age, *given that an MCV2 dose is received*." Since receiving MCV2 guarantees having received MCV1, receiving MCV2 influences the likelihood of prior immunity. Specifically, someone who receives an MCV2 dose is *more likely* than the average person to have prior immunity from MCV1 at MCV2 age, meaning (9) **overestimates** MCV2-derived immunity. For MCV1 and SIAs, we assume that the likelihood of receiving a dose was independent from the likelihood of receiving other doses, so this issue does not arise.

Derivation of the updated formula. We seek $P(s = \text{MCV2} \mid t)$, the probability of MCV2-derived immunity for an individual born in semimonth t and year b. This requires three conditions: MCV2 dose received, MCV2 dose seroconverts, and no prior immunity:

$$P(s = \text{MCV2} \mid t) = \varepsilon_2 \cdot P(\text{MCV2 received} \cap \text{no prior immunity}). \tag{10}$$

The original formulation implicitly assumed that "MCV2 dose received" and "no prior immunity" are independent. However, as explained above, these events are *not* independent, and

$$P(\text{no prior immunity} \mid \text{MCV2 received}) < P(\text{no prior immunity}). \tag{11}$$

The correct factoring is therefore as follows:

$$P(\text{MCV2 received} \cap \text{no prior immunity}) = P(\text{no prior immunity} \mid \text{MCV2 received}) \cdot C_2(t). \tag{12}$$

We can break down $P(\text{no prior immunity} \mid \text{MCV2 received})$ as follows:

$$\begin{aligned}
&P(\text{no prior immunity} \mid \text{MCV2 received}) && (13)\\
&\quad = P(\text{not inf. by } a_{\text{MCV2}} \cap \text{not imm. by MCV1} \cap \text{not imm. by SIA} \mid \text{MCV2 received}) && (14)\\
&\quad = P(\text{not imm. by MCV1} \mid \text{not inf. by } a_{\text{MCV2}}, \text{ not imm. by SIA, MCV2 received}) && (15)\\
&\qquad * \, p(\text{not inf. by } a_{\text{MCV2}} \cap \text{not imm. by SIA} \mid \text{MCV2 received}) && (16)\\
&\quad = (1 - \varepsilon_1)\left(1 - F(a_{\text{MCV2}}, b) - V_{\text{cum}}^{\text{SIA}}(t)\right) && (17)
\end{aligned}$$

Combining (10)–(17):

$$\boxed{P(s = \text{MCV2} \mid t) = \varepsilon_2 \, C_2(t) \, (1 - \varepsilon_1) \left(1 - F(a_{\text{MCV2}}, b) - V_{\text{cum}}^{\text{SIA}}(t)\right).} \tag{18}$$

Assumption: MCV2 is a second dose

In this new MCV2 formulation, we assume that receiving MCV2 *guarantees* having received MCV1. The original formulation assumed that receiving MCV2 has *no correlation* with receiving MCV1. While our new assumption seems more permissible than the original assumption, it is still worth questioning, particularly because it could be a convention to mark down any dose received around 15 months of age as "MCV2," even if no prior dose was received. We plan to do some research to investigate this.

6 Estimation of $P_b(I)$: the full-immunity normalization

In this section, we explain how to estimate $P_b(I)$ using a normalization method introduced in Thakkar et al. [2025], although we have made no methodological changes to the approach. The model assumes that every individual in cohort b eventually acquires immunity from some source:

$$V(b;\, P_b(I)) + P_b(I) = 1, \qquad V(b;\, P_b(I)) = \sum_{s'} P(s = s' \mid b), \tag{19}$$

where the sum is over all vaccine-derived sources (MCV1, MCV2, and all SIAs). Recall that the probabilities $P(s = s' \mid b)$ depend on $P_b(I)$ via $F(a_{\text{sm}}, b)$ (see (4) and Subsection 4.4), so $V(b;\, P_b(I))$ depends on $P_b(I)$. This interdependence initially makes (19) look complicated.

However, we will see that $V(b; P_b(I))$ is linear in $P_b(I)$, and this will allow us to solve for the unique $P_b(I)$ value satisfying $V(b;\, P_b(I)) + P_b(I) = 1$.

Assumption: Immunity is obtained by age 25 and is lifetime durable

We assume that everyone derives immunity by age A (i.e., that those who are not vaccinated by age A get infected by age A). We take $A = 25$; observed age-at-infection distributions suggest that a negligible portion of cases occur above age 25. We also assume that immunity is lifetime durable. This is a very measles-specific assumption (and could be false, but in ways that don't appear to significantly influence transmission). That being said, it could be worth thinking about how a biologically known rate of immunity waning could be incorporated, which could be a step towards extending this immunity profile method beyond measles.

6.1 Linearity in $P_b(I)$

A key structural property is that $V(b;\, P_b(I))$ is *linear* in $P_b(I)$, regardless of the chronological ordering of vaccination events. To see this, recall that each vaccination event s' computes new immunity $P(s = s' \mid t)$ as a constant (efficacy $\times$ coverage) multiplied by a susceptible fraction. The susceptible fraction is $1 - F(a_{\text{sm}}, b) - V_{\text{cum}}(t)$ (or $(1-\varepsilon_1)(1 - F(a_{\text{MCV2}}, b) - V_{\text{cum}}^{\text{SIA}}(t)$ in the case of MCV2). From (4), we know that $F(a_{\text{sm}}, b)$ is linear in $P_b(I)$. Since $V_{\text{cum}}(t)$ is initialized to 0 and is then a cumulative sum of these "new immunity" terms, we see by an inductive argument that $V_{\text{cum}}(t)$ (and hence $P(s = s' \mid t)$) always remains linear in $P_b(I)$. Finally, since the aggregation from $P(s = s' \mid t)$ to $P(s = s' \mid b)$ is linear in $P(s = s' \mid t)$, we see that $P(s = s' \mid b)$ and $V(b;\, P_b(I))$ are linear in $P_b(I)$.

6.2 Exact two-point solve

Since $f(P_b(I)) = V(b;\, P_b(I)) + P_b(I)(b) - 1$ is linear in $P_b(I)$, evaluating at two points gives the exact root:

$$\hat{P}_b(I) = \frac{-f(0)\ \cdot\ P_b(I)^{\text{right}}}{f(P_b(I)^{\text{right}}) - f(0)}, \tag{20}$$

where $P_b(I)^{\text{right}} = 0.1$ (or any other nonzero value).

7 From cohort profiles to year-level incidence

The unnormalized expected incidence at age a in calendar year y is:

$$\hat{I}(a,\, y) = B(y-a)\ \hat{P}_{b=y-a}(I)\ \hat{P}_{b=y-a}(a \mid I), \tag{21}$$

Each age a within a fixed year y references a different cohort: the year-level distribution stitches together one cell from each of many cohort distributions. The model-implied year-based age distribution is obtained by normalizing:

$$\hat{P}_y(a \mid I) = \frac{\hat{I}(a,\, y)}{\sum_{a'=0}^{A} \hat{I}(a',\, y)}. \tag{22}$$

8 Reporting rate estimation

For a fixed year y, summing $\hat{I}(a, y)$ over the possible ages gives the total expected infections in year y:

$$\hat{I}(y) = \sum_{a=0}^{A} \hat{I}(a, y). \tag{23}$$

Comparing this to the number n_y of observed cases in year y gives an estimate for the reporting rate in year y:

$$\hat{r}(y) \approx \frac{n_y}{\hat{I}(y)}. \tag{24}$$

Future work: age-specific reporting rates

We could look at $n_y(a)/\hat{I}(a, y)$ to get age-specific reporting rate estimates. However, the way in which we've estimated $\hat{I}(a, y)$ from the cohort age-at-infection distributions implicitly assumes that there's no difference in the reporting rate across ages, so $n_y(a)/\hat{I}(a, y)$ probably wouldn't currently be meaningful. But this could be given more thought.

References

Niket Thakkar, Avuwa Joseph Oteri, and Kevin McCarthy. Routine immunization intensification, vaccination campaigns, and measles transmission in Southern Nigeria. *medRxiv*, 2025. doi: https://doi.org/10.1101/2025.02.24.25322796